

Tuan Nguyen

TECH LEAD

Hanoi, Vietnam

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Summary

I design and build complex systems, with a 12-year track record spanning advertising, live-streaming, enterprise chat system, and a full-featured enterprise agent platform. Parallel to this, my passion for independent research has led me to explore everything from visual search, neural vocoder to generative AI editing. As a product-focused engineer ready to lead, my goal is to channel this unique combination of architectural discipline and deep research into building great things.

Skills

Back-end	Golang, Node.Js, Python, REST API, System Design, Live-streaming, System design
Front-end	React, Redux
Database	Mysql, Postgres, Redis, Cassandra
Machine Learning	Pytorch, Numpy
Dev-Ops	Docker, Gitlab CI
Programming	Golang, Javascript/Typescript, Python, Java
Languages	English, Vietnamese

Work Experience

Vinbase

Hanoi, Vietnam

SOFTWARE ENGINEER

May, 2023 - Now

- Drove the technical design of the company's next gen LLM-based agent platform, architecting it from an open-source core into a full-featured product through 50+ authored proposals.
- Engineered the platform's core AI modules with a focus on thoughtful design, creating a advanced RAG system, scalable agentic workflows, and a comprehensive observability tools.
- Championed a superior developer and user experience (DX/UX), boosting team productivity 3x by creating a unified, highly reusable library and modernizing the development stack.
- Hardened the platform's legacy (intent-based) system for key enterprise clients, improving its stability and maintainability through methodical refactoring and a significant increase in test coverage.

Gapo

Hanoi, Vietnam

SOFTWARE ENGINEER

May, 2022 - 2023

- Developed and maintained backend system for chat and bot features for **Gapowork**, a digital workspace for companies and organizations. Gapowork won **Outstanding Digital Solution Award** at Vietnam Digital Awards 2021 and are widely used by many large brands in Vietnam such as BIDV, Yody, Momo,...
- In addition to developing new APIs, I optimized our system for better performance and reliability, resulting in a 100-200% improvement in p99 latency and high availability. This significantly enhanced our users' experience, as they rely on our app daily.

Cortexapp

Hanoi, Vietnam

FOUNDER

2021 - 2022

- I built an Airtable-like product and then a Personal Knowledge System.
- However, I stopped working on this product because I don't believe these approaches are effective in solving people's problems.

Umbala Technology

Hanoi, Vietnam

TECHNICAL LEADER

Jun. 2018 - Feb. 2021

- Worked with other developers to build API endpoints for **Umbala.Tv** app, a trending, life-style live-shopping platform. Umbala.Tv is a full functional e-commerce platform.
- Worked with other team members to build webapps, merchant dashboard, admin dashboard for Umbala.Tv using React technology.
- I built a p2p streaming protocol (you can think it as Torrent but for live stream). POC is [here](#).
- My team of five built a scalable video streaming platform to serve streaming for app. The system was built from scratch and can deliver low-latency streaming (<3s) with a peer-to-peer (P2P) feature to cut off CDN load upto 60%. Early video demo can be found [here](#).

Admicro, VCcorp.

Hanoi, Vietnam

TECHNICAL TEAM LEADER

Nov. 2017 - Feb. 2018

- Built and scaled Admicro's advertising system, which handles more than 600M ad requests per day, making it one of the largest ad systems in Vietnam.
- Led a team of engineers responsible for building ad services and many data systems to handle all things behind the scene.

Admicro, VCcorp.

ASSOCIATIVE TEAM LEADER

Hanoi, Vietnam

Apr. 2017 - Nov. 2017

Admicro, VCcorp.

BIGDATA ENGINEER

Hanoi, Vietnam

Aug. 2014 - Apr. 2017

- Designed and implemented the first frequency capping system using Spark Streaming and Cassandra.
- Proposed and implemented new solution to detect user location from IP in Vietnam with much better accuracy than existing paid solutions in the market.
- Built near real-time recommendation system for dynamic ads product, which allows to display ads related to their recently visited product pages.

Sunnet

HTML5 DEVELOPER INTERN

Hanoi, Vietnam

Jul. 2012 - June. 2013

- Researched and developed web games using HTML5 technologies. I developed 5 games using different technologies such as RaphaelJS, Box2D.js, TweenJS, Three.js,...

Honors & Awards

Best Employee Award

VCCorp

2015, 2016, 2017

Side Projects

obsidian-template

OBSIDIAN TEMPLATE

2022

- My personal obsidian templates. Obsidian is my favorite note taking app. I use it to write down ideas and what I have learned.

linearapp_clone

A LINEAR APP CLONE WITH REACT AND TAILWIND CSS

2021

- A quick **Linear.app** clone built with React and Tailwind CSS in 1 week. This project became the top of the day ReactJS's reddit channel.

pytorch-spleeter

A PYTORCH IMPLEMENTATION OF SPLEETER

2020

- Implementation of **Spleeter**, a popular music source separation library, in Pytorch. It can load 2-stem model to separate voice from instruments.

cnn_vocoder

A CNN-BASED VOCODER

2019

- An experimental CNN based neural vocoder, which generates speech from Spectrogram. It's in my attempt to build much faster vocoder than WaveNet. It was inspired by m-cnn architecture.

visual_search

A VISUAL SEARCH ENGINE BASED ON ELASTICSEARCH AND TENSORFLOW

Feb 2017 - March 2017

- A visual search engine like Pinterest, based on Elasticsearch and Tensorflow, which may be one of the first open source Deep Learning-based visual search engines.
- I built this in two weeks of hacking, trying to build a visual search based on what I had learned.
- This project was featured on Github's Trending.

Publications

KSE

MAIN AUTHOR

2015

- N. D. Tuan, N. Q. Manh, D. V. Sang, Huynh Thi Thanh Binh and N. T. Thuy, "A Study on Non-sparse Dictionary Learning for Pattern Classification," 2015 Seventh International Conference on Knowledge and Systems Engineering (KSE), Ho Chi Minh City, Vietnam, 2015, pp. 371-376, doi: 10.1109/KSE.2015.66.
- N. Q. Manh, N. D. Tuan, D. V. Sang, H. T. T. Binh and N. T. Thuy, "Uniform Detection in Social Image Streams," 2015 Seventh International Conference on Knowledge and Systems Engineering (KSE), Ho Chi Minh City, Vietnam, 2015, pp. 180-185, doi: 10.1109/KSE.2015.63.

Education

Hanoi University Of Science And Technology

Hanoi, Vietnam

ENGINEER DEGREE. IN COMPUTER SCIENCE

2010-2015

- GPA: 3.52/4 (8.8/10).